

AWS Security Group Monitoring Lab

CloudTrail • CloudWatch • SNS • SQS • Splunk

Introduction

This project demonstrates real-time Security Group monitoring and detection in Amazon Web Services using CloudTrail, CloudWatch, SNS, SQS, and Splunk. The objective is to detect and monitor Security Group activities such as rule additions, rule removals, Security Group creation, and deletion events.

AWS Security Groups act as virtual firewalls that control inbound and outbound traffic for EC2 instances. Misconfigured or publicly exposed Security Groups can lead to unauthorized access, infrastructure compromise, and cloud security incidents. To address this risk, monitoring and alerting mechanisms were implemented using AWS native services and SIEM integration.

The project captures Security Group events from CloudTrail, processes them through CloudWatch Metric Filters and Alarms, sends notifications using SNS and SQS, and forwards events to Splunk for centralized monitoring and detection engineering.

Monitored Security Group Events:

- CreateSecurityGroup
- AuthorizeSecurityGroupIngress
- RevokeSecurityGroupIngress
- DeleteSecurityGroup

This project demonstrates practical cloud security monitoring, detection engineering, alerting workflows, and SIEM integration in a real-world SOC-style environment.

Step 1 — Launch EC2 Instance

First, launch an EC2 instance that will serve as the target for Security Group activity monitoring.

Navigation:

- Go to EC2
- Click Launch Instance
- Configure and launch your instance

EC2 > Instances > Launch an instance

Launch an instance

Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Info

Name

SOC-Monitoring-Server

Add additional tags

Application and OS Images (Amazon Machine Image)

Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI

ami-0f5b1543e7934f48 (64-bit x86), uefi-preferred / ami-0419317e14b33ad8e (64-bit ARM), uefi

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Summary

Info

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11...[read more](#)

ami-0f5b1543e7934f48

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

EC2 > Instances > Launch an instance

Instance type

Info | Get advice

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0133 USD per Hour On-Demand Windows base pricing: 3.0194 USD per Hour

Free tier eligible

All generations

Compare instance types

Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

SOC-Monitoring-Servers

Create new key pair

Network settings

Info

Network

vpc-0e072a721168463ab

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Firewall (security groups)

Info

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

Summary

Info

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11...[read more](#)

ami-0f5b1543e7934f48

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

EC2 > Instances > Launch an instance

Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

Allow HTTPS traffic from the Internet

To set up an endpoint, for example when creating a web server

Allow HTTP traffic from the Internet

To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage

Info

Advanced

1x 8 GiB gp3

Root volume, 3000 IOPS, Not encrypted

Add new volume

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

S3 Files - new

EFS

FSx

None

Advanced details

Info

Summary

Info

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11...[read more](#)

ami-0f5b1543e7934f48

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

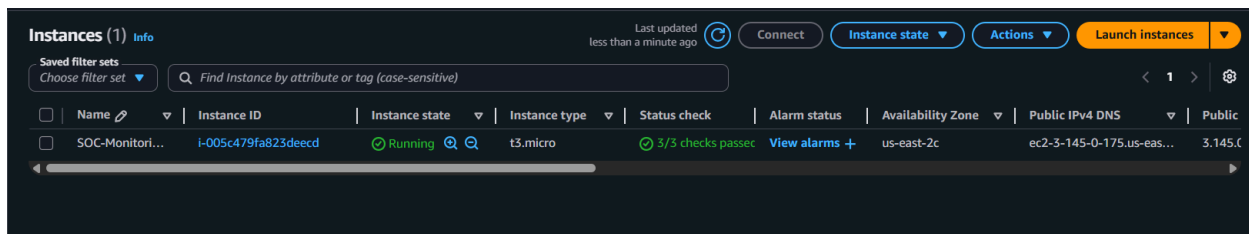
Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code



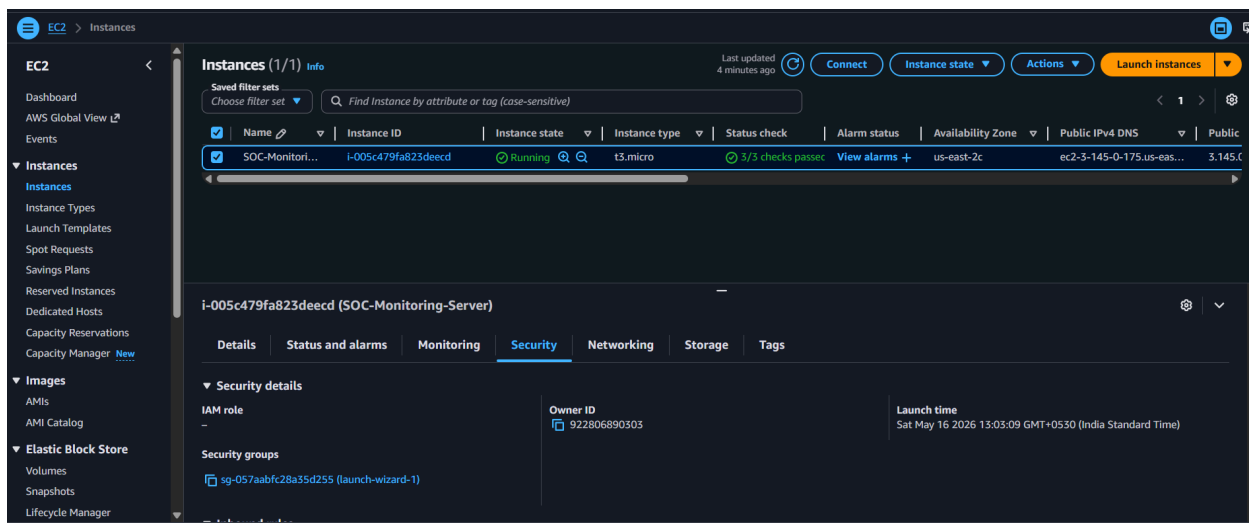
Step 2 — Identify Security Group ID

After the instance is launched, retrieve the Security Group ID. You will use this ID when testing and verifying detection events.

Navigation:

- Go to EC2 → Instances → Security tab
- Copy the Security Group ID (format: sg-xxxxxxx)

sg-xxxxxxx



Step 3 — Simulate Security Group Attack Activity

Generate detection events by modifying Security Group rules. This simulates attacker behaviour that SOC analysts monitor for.

Navigation:

- Go to EC2 → Security Groups
- Select your Security Group

Add:

Type	Port	Source
SSH	22	0.0.0.0/0

Step 4 — Edit Inbound Rules

Navigate to the inbound rules editor to add a potentially dangerous rule that will be captured by CloudTrail.

Action: Click Edit inbound rules

EC2

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

sg-057aabfc28a35d255 - launch-wizard-1

Details

Security group name

launch-wizard-1

Security group ID

sg-057aabfc28a35d255

Description

launch-wizard-1 created 2026-05-16T07:30:19.656Z

VPC ID

vpc-0e072a721168463ab

Owner

922806890303

Inbound rules count

3 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (3)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-03173459c766f7695	IPv4	HTTPS	TCP	443	0.0.0.0/0
-	sgr-07e630ee83c0d577b	IPv4	HTTP	TCP	80	0.0.0.0/0
-	sgr-006fe737cf4d4515e0	IPv4	SSH	TCP	22	0.0.0.0/0

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID

sg-03173459c766f7695

Type

HTTPS

Protocol

TCP

Port range

443

Source

Custom

Description - optional

Delete

sg-07e630ee83c0d577b

HTTP

TCP

80

Custom

0.0.0.0/0

0.0.0.0/0

Delete

sg-006fe737cf4d4515e0

SSH

TCP

22

Custom

0.0.0.0/0

0.0.0.0/0

Delete

-

HTTPS

TCP

443

Anywhere...

0.0.0.0/0

0.0.0.0/0

Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

Preview changes

Save rules

Step 5 — Create a Dangerous Rule

Add a new inbound rule that exposes a port publicly. This action is intentionally misconfigured to simulate a cloud security incident.

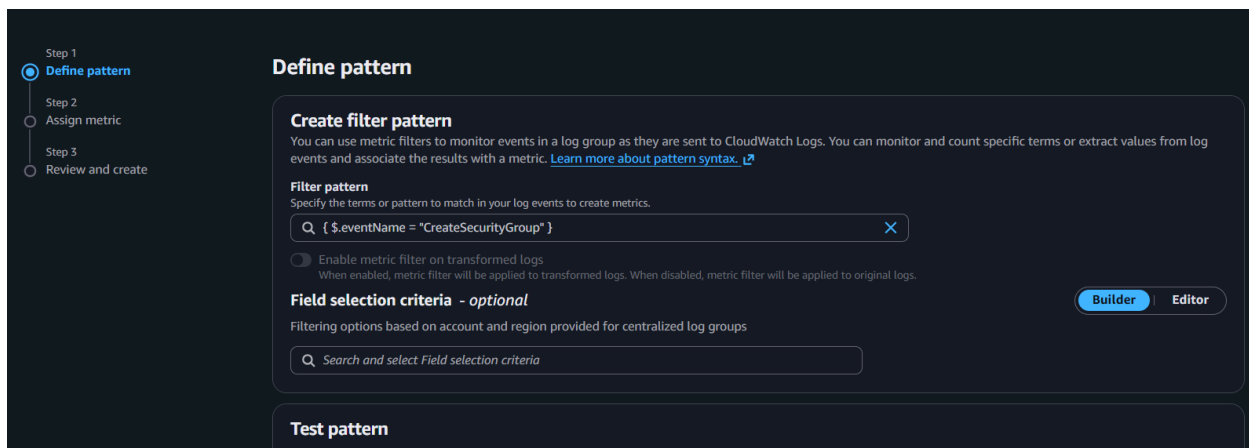
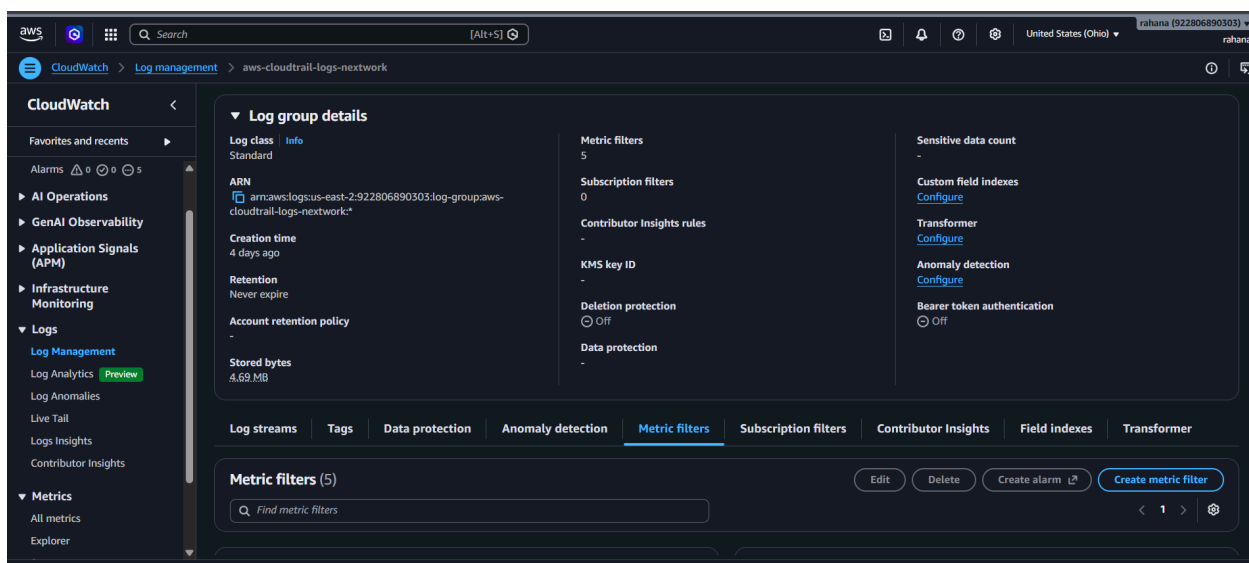
Action: Add the new inbound rule, then click Save rules

What Happens Internally

When you save the inbound rule, AWS automatically records a CloudTrail API event:

AuthorizeSecurityGroupIngress

This is the primary event detected by the monitoring pipeline. CloudTrail logs it within seconds of the rule change.



Define pattern

Step 2

Assign metric

Step 3

Review and create

Create filter name

Log events that match the pattern you define are recorded to the metric that you specify. You can graph the metric and set alarms to notify you.

Filter name

Create-SecurityGroup-Alert

Filter pattern

{ \$.eventName = "CreateSecurityGroup" }

☐ Enable metric filter on transformed logs
When enabled, metric filter will be applied to transformed logs. When disabled, metric filter will be applied to original logs.

Metric details

Metric namespace
Namespaces let you group similar metrics. [Learn more](#)

SecurityMonitoring ☒ Create new

Namespaces can be up to 255 characters long; all characters are valid except for colon(:) at the start of the name.

Metric name
Metric name identifies this metric, and must be unique within the namespace. [Learn more](#)

CreateSecurityGroupEvent

Metric name can be up to 255 characters long; all characters are valid except for colon(:), asterisk(*), dollar(\$), and space().

Metric value
Metric value is the value published to the metric name when a Filter Pattern match occurs.

1

Valid metric values are: floating point number (1, 99.9, etc.), numeric field identifiers (\$1, \$2, etc.), or named field identifiers (e.g. \$requestSize for delimited filter pattern or \$.status for JSON-based filter pattern - dollar (\$) or dollar dot (\$.) followed by alphanumeric and/or underscore (_) characters).

Step 1

Define pattern

Step 2

Assign metric

Step 3

Review and confirm

Step 1: Pattern

Create filter pattern

Filter pattern

{ \$.eventName = "CreateSecurityGroup" }

[Edit](#)

Step 2: Metric

Assign metric

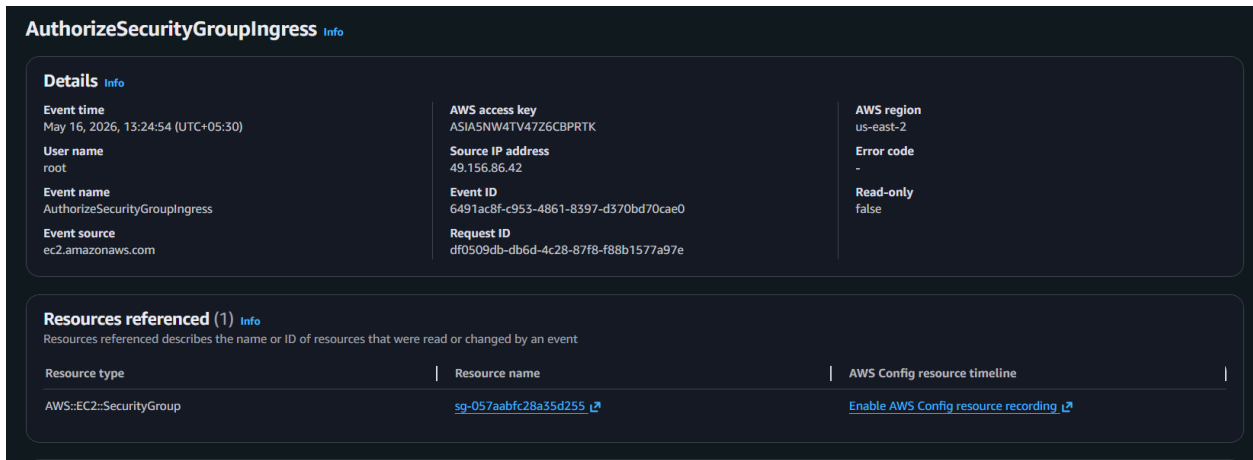
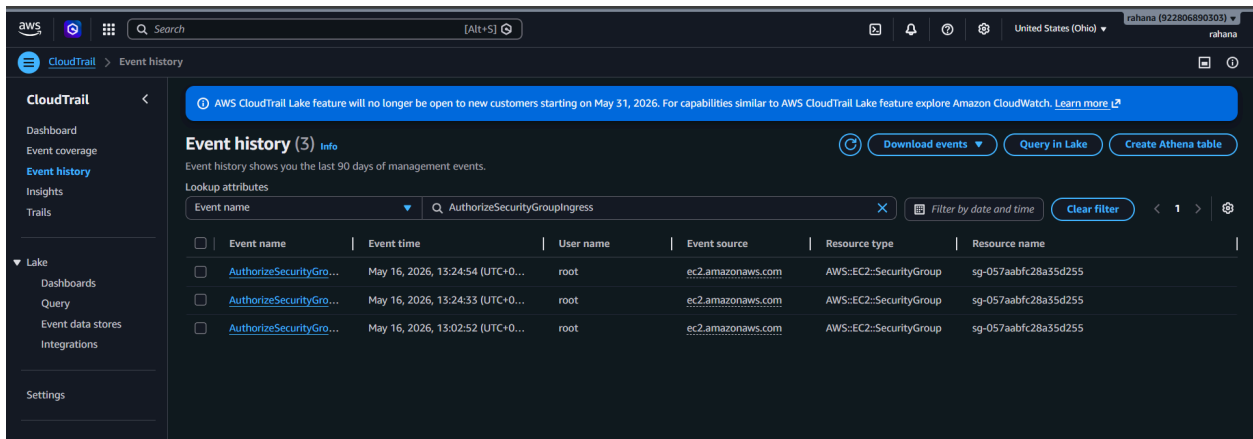
Filter name Create-SecurityGroup-Alert	Metric name CreateSecurityGroupEvent
Metric namespace SecurityMonitoring	Applied on transformed logs ☒
Metric value 1	Default value -
Unit -	

Step 6 — Verify Event in CloudTrail

Confirm that CloudTrail has captured the Security Group modification event before proceeding to configure CloudWatch alerts.

Navigation:

- Go to CloudTrail → Event History
- Search for: AuthorizeSecurityGroupIngress
- Open the event to inspect the full API call details



Security Group Events to Monitor	
Event Name	Purpose
CreateSecurityGroup	Detect SG creation
AuthorizeSecurityGroupIngress	Detect rule addition
RevokeSecurityGroupIngress	Detect rule removal
DeleteSecurityGroup	Detect SG deletion

Create CloudWatch Metric Filters

CloudWatch Metric Filters parse CloudTrail log events and convert them into quantifiable metrics. Configure one filter per Security Group event type.

Navigation:

- Go to CloudWatch → Log Groups
- Open your CloudTrail Log Group
- Click Create Metric Filter

Step-by-Step: Metric Filter for Each Event

1. CreateSecurityGroup

2. AuthorizeSecurityGroupIngress

Section	Field	Value to Fill
Log Group Selection	Log Group	CloudTrail Log Group
Metric Filter	Filter Pattern	<code>{ \$.eventName = "AuthorizeSecurityGroupIngress" }</code>
Metric Details	Filter Name	<code>AuthorizeSecurityGroupIngressFilter</code>
Metric Details	Metric Namespace	<code>SecurityMonitoring</code>
Metric Details	Metric Name	<code>AuthorizeSecurityGroupIngressEvent</code>
Metric Details	Metric Value	<code>1</code>
Default Value	Default Value	<code>0</code>
Unit (Optional)	Unit	<code>Count</code>
Create Action	Final Step	Click <code>Create Metric Filter</code>

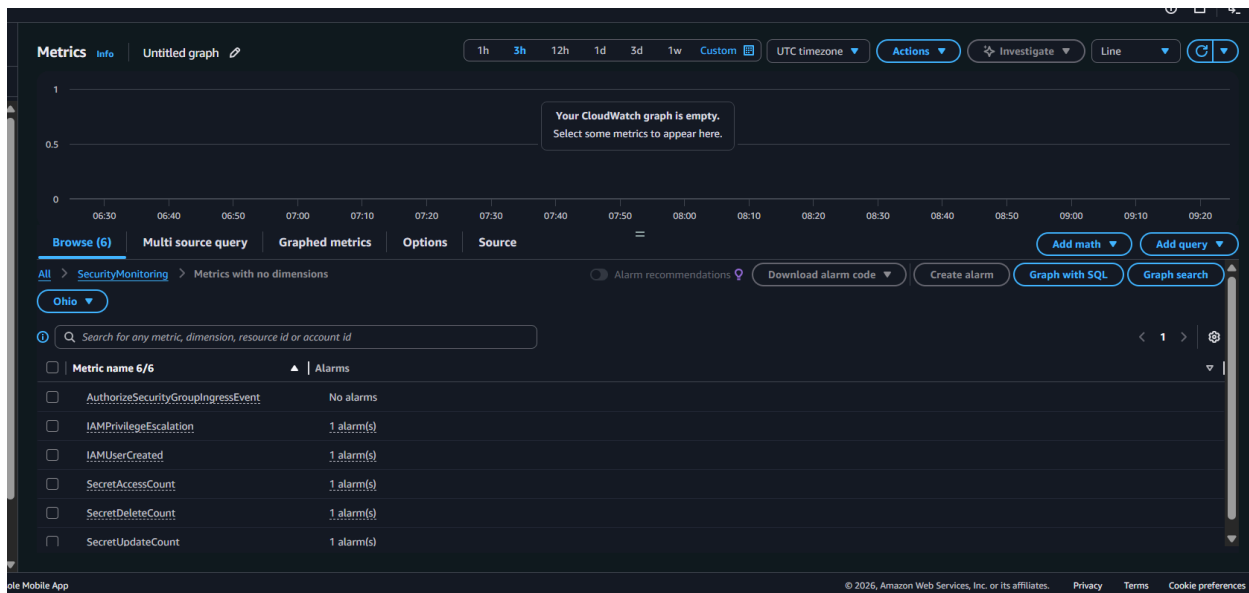
3. RevokeSecurityGroupIngress

Section	Field	Value to Fill
Log Group Selection	Log Group	CloudTrail Log Group
Metric Filter	Filter Pattern	<code>{ \$.eventName = "RevokeSecurityGroupIngress" }</code>
Metric Details	Filter Name	RevokeSecurityGroupIngressFilter
Metric Details	Metric Namespace	SecurityMonitoring
Metric Details	Metric Name	RevokeSecurityGroupIngressEvent
Metric Details	Metric Value	1
Default Value	Default Value	0
Unit (Optional)	Unit	Count
Create Action	Final Step	Click <code>Create Metric Filter</code>

4. DeleteSecurityGroup

Section	Field	Value to Fill
Log Group Selection	Log Group	CloudTrail Log Group
Metric Filter	Filter Pattern	<code>{ \$.eventName = "DeleteSecurityGroup" }</code>
Metric Details	Filter Name	DeleteSecurityGroupFilter
Metric Details	Metric Namespace	SecurityMonitoring
Metric Details	Metric Name	DeleteSecurityGroupEvent
Metric Details	Metric Value	1
Default Value	Default Value	0
Unit (Optional)	Unit	Count
Create Action	Final Step	Click <code>Create Metric Filter</code>

After creating the metric filters, generate test events (add or remove inbound rules) and verify they appear in CloudTrail Event History before proceeding.



Create CloudWatch Alarms

Configure CloudWatch Alarms on each metric filter to trigger SNS notifications when Security Group events are detected.

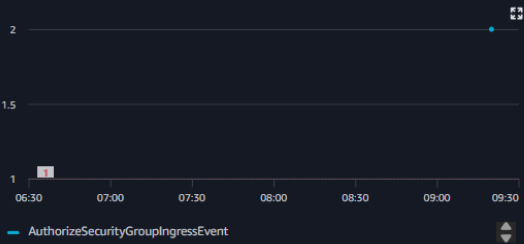
Navigation:

- Go to CloudWatch → All Metrics → Security Monitoring
- Select the event metric
- Click Create Alarm

Metric

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.



AuthorizeSecurityGroupIngressEvent

Namespace
SecurityMonitoring

Metric name
AuthorizeSecurityGroupIngressEvent

Statistic
Sum

Period
5 minutes

Conditions

Threshold type

☒ **Static**
Use a value as a threshold

☐ **Anomaly detection**
Use a band as a threshold

Whenever AuthorizeSecurityGroupIngressEvent is...
Define the alarm condition.

☐ Greater
> threshold

☒ **Greater/Equal**
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...
Define the threshold value.

1

Must be a number.

► **Additional configuration**

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

SecurityAlertsbynextwork

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)
nadi.lan23@lpu.in - View in SNS Console

Add notification

Lambda action

Add Lambda action

Auto Scaling action

Add Auto Scaling action

Create Alarm for Each Event

1. CreateSecurityGroup

Add alarm details

Name and description

Alarm name

SecurityGroup-Ingress-Modification-Alert

Alarm description - optional [View formatting guidelines](#)

Edit

Preview

This is an H1

double asterisks will produce strong character

This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024).

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Tags - optional [Info](#)

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel

Previous

Next

2. AuthorizeSecurityGroupIngress

Alarm Configuration Table		
Section	Field	Value
Metric Selection	Namespace	SecurityMonitoring
Metric Selection	Metric Name	AuthorizeSecurityGroupIngressEvent
Metric Selection	Statistic	Sum
Metric Selection	Period	5 Minutes
Conditions	Threshold Type	Static
Conditions	Whenever Metric Is	Greater/Equal
Conditions	Than	1
Notification	Alarm State Trigger	In Alarm
Notification	SNS Topic	SecurityAlerts
Alarm Details	Alarm Name	SecurityGroup-Ingress-Modification-Alert
Alarm Details	Alarm Description	Detects Security Group inbound rule additions

3. RevokeSecurityGroupIngress

Alarm Configuration Table

Section	Field	Value
Metric Selection	Namespace	SecurityMonitoring
Metric Selection	Metric Name	RevokeSecurityGroupIngressEvent
Metric Selection	Statistic	Sum
Metric Selection	Period	5 Minutes
Conditions	Threshold Type	Static
Conditions	Whenever Metric Is	Greater/Equal
Conditions	Than	1
Notification	Alarm State Trigger	In Alarm
Notification	SNS Topic	SecurityAlerts
Alarm Details	Alarm Name	SecurityGroup-Rule-Removal-Alert
Alarm Details	Alarm Description	Detects removal of Security Group inbound rules

4. DeleteSecurityGroup

Alarm Configuration Table

Section	Field	Value
Metric Selection	Namespace	SecurityMonitoring
Metric Selection	Metric Name	DeleteSecurityGroupEvent
Metric Selection	Statistic	Sum
Metric Selection	Period	5 Minutes
Conditions	Threshold Type	Static
Conditions	Whenever Metric Is	Greater/Equal
Conditions	Than	1
Notification	Alarm State Trigger	In Alarm
Notification	SNS Topic	SecurityAlerts
Alarm Details	Alarm Name	SecurityGroup-Deletion-Alert
Alarm Details	Alarm Description	Detects deletion of AWS Security Groups

Final Verification — Generate Event & Confirm Alert

Perform an end-to-end test to confirm the full monitoring pipeline is working: from Security Group change through CloudTrail, CloudWatch, SNS notification, and Splunk SIEM ingestion.

Step 1 — Generate Test Event

Navigation:

- Go to EC2 → Security Groups
- Select your Security Group
- Click Edit inbound rules
- Add a new rule
- Click Save rules

This generates an AuthorizeSecurityGroupIngress event in CloudTrail.

Step 2 — Wait 1–5 Minutes

AWS services require a brief processing window before the alert propagates through the pipeline:

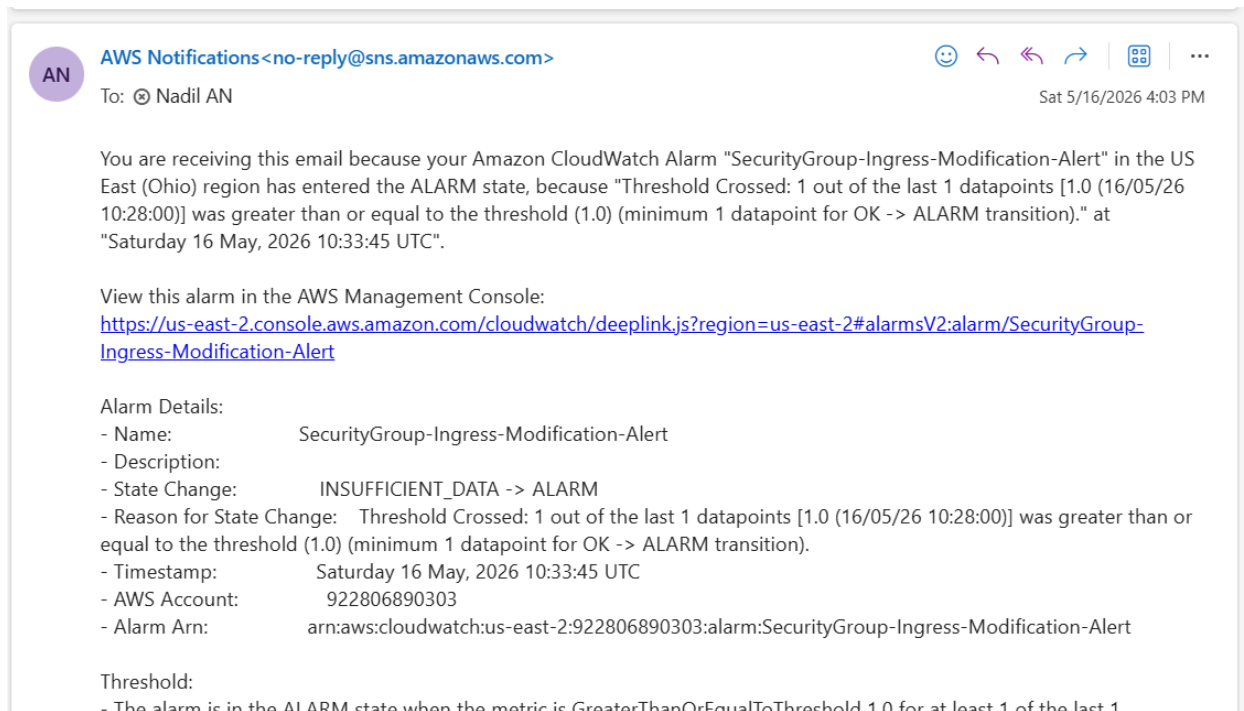
- CloudTrail: log processing
- CloudWatch: metric evaluation
- SNS: notification dispatch

- Splunk: log ingestion and indexing

Step 3 — Verify Email Notification

Check your email inbox for the SNS alarm notification. It will look similar to:

ALARM: "SecurityGroup-Ingress-Modification-Alert" in US East (Ohio)



Conclusion

In this project, a complete AWS Security Group monitoring and alerting pipeline was successfully implemented using CloudTrail, CloudWatch, SNS, SQS, and Splunk SIEM integration. Security Group activities such as inbound rule additions, rule removals, Security Group creation, and deletion events were captured and monitored in real time.

CloudTrail recorded AWS API activities while CloudWatch Metric Filters and Alarms were configured to detect specific Security Group events. SNS notifications and SQS integration enabled automated alert forwarding, and Splunk provided centralized log analysis and detection monitoring.

Key Outcomes:

- Real-time cloud security monitoring
- Security event detection across four event types
- AWS-native alerting via CloudWatch, SNS, and SQS
- SIEM integration with Splunk for centralized log analysis
- Detection engineering workflows validated end-to-end